MIL-I-60642(MU)
24 August 1967

SUPERSEDING
FA-PD-MI-2583, Rev. 0
3 May 1963

MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M80 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, M80, Assembly, loaded with Cartridge, Initiator, M69, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MS28741	- Hose Assembly, Detachable End-Fitting, Medium Pressure

DRAWINGS

Munitions Command

F10522664	- Initiator, Cartridge Actuated, M80, Assembly
C8797660	- Box, Packing, Ammunition for Initiator, Cartridge Actuated, M9 or M80 in Fiber Container, M381

FSC 1377

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2.1 DRAWINGS (Cont'd)

Munitions Command
C8797661 - Container, Ammunition, Fiber M381 for Initiator, Cartridge Actuated, M9 or M80
IEL-10522664 - Inspection Equipment List for Initiator, Cartridge Actuated, M80 Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Torque.

3.2.1 Assembly.- An application of 400 plus 20 inch-pounds of seating torque shall be applied to the final assembly of the initiator cap and body.

3.2.2 Seating.- The body, cap and filter shall be assembled in accordance with the applicable drawing. A seating torque of 300 + 100 inch-pounds shall be applied to the final assembly of the cap and body.

3.2.3 Breakaway.- A breakaway torque of 400 inch-pounds shall be applied between the initiator cap and body and between the head and body.

3.3 Residual magnetism.- The initiator shall not deflect the indicator of the compass by more than 3° in either direction when passing in front of the compass at a distance of 6 inches.

3.4 Functional

3.4.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be produced at the end of 30 feet of aircraft hose, size No. 4, as specified in MS28741.

TABLE I

Temperature	Pressure (PSI) Min.	Max.	Ignition Delay Max. (See 6.2)
-65° to -70°F	1000	3000	65 milliseconds
65° to 75°F	1000	3000	65 milliseconds
160° to 165°F	1000	3000	65 milliseconds

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3.4.1.1 Initiation.- The initiator shall be fired by a source delivering an output equal to that of an M3A1 initiator at the end of 15 feet of aircraft hose, size No. 4, in accordance with MS28741, at 70° ± 5°F.

3.4.2 Carbon monoxide gas leakage.- Initiators loaded in the accepted cartridges and temperature conditioned at 65° - 5°F, 70° + 5°F and 165° + 5°F, when fired into a sealed total volume of 5.9 cubic inches shall not produce carbon monoxide gas leakage greater than 5.0 cubic centimeters total. Volume of gas leakage shall be recorded at 10 minutes and 60 minutes after firing.

3.4.3 The initiator shall not fail to fire.

3.4.4 The initiator shall function without mechanical failure (see 6.1).

3.5 Workmanship.- Requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- The component shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

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4.2.1 Lot

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Initiator, M69, Drawing D8593496, covering a detail of Drawing F10522664

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. "O" ring missing	Visual
4. Sealing disc missing, damaged or improperly secured	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. "O" ring split, cut or cracked	Visual
102. Foreign matter	Visual

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4.2.2.1 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Case scratched or dented	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Drawing F10522664, less Drawings D8593496, D10522666, D10522667, D10522668, C10522670 and C10522671

CRITICAL	
1. Firing pin missing	Visual
2. Firing pin incorrectly assembled	Visual
3. Shear pin missing	Visual
4. "O" ring missing	Visual
MAJOR	
101. Threads damaged	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
201. "O" ring not lubricated	Visual

4.2.2.3 Assembly, Drawing F10522664, less Drawings F10522493, B10522949, C10522495, C10522670 and C10522671

CRITICAL	
1. Filter missing or improperly assembled	Visual
2. Cartridge missing	Visual
3. Cartridge loose	Visual and Manual
4. "O" ring missing	Visual
MAJOR	
101. Any part missing, except as otherwise classified	Visual
102. Thread damaged	Visual
103. Foreign matter	Visual

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4.2.2.3 (Cont'd)

Defect	Method of Inspection
MINOR	
201. "O" ring not lubricated	Visual

4.2.2.4 Assembly, Drawing F10522664 less C10522670 and C10522671

CRITICAL	
1. Any part missing, except as otherwise classified	Visual
2. Marking missing, incorrect or illegible	Visual
3. Foreign matter in gas port	Visual
MAJOR	
101. Thread damaged	Visual
MINOR	
201. Location of marking incorrect	Visual
202. Sealing compound missing from set screws	Visual
203. Copalite missing from body thread	Visual
204. Shipping plug or cap missing	Visual
205. Protective finish inadequate or defective	Visual
206. Burr	Visual
207. Foreign matter	Visual

4.2.2.5 Assembly, Drawing F10522664

CRITICAL	
None defined	
MAJOR	
101. Assembly of block and strap incorrect or insecure	Visual and Manual
MINOR	
None defined	

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4.2.2.6 Packing in Fiber Container, Drawing C8797661

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.7 Packing in Wooden Box, Drawing C8797660

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Visual and Manual
202. Hardware missing, loose, broken or improperly engaged	Visual and Manual
203. Strapping missing, loose, broken or improperly assembled	Visual and Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

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4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- The threaded joint of the initiator cap body and body-cap (end) of each initiator shall be subjected to the breakaway torque application. Failure of any initiator to comply with the requirements of 3.2.3 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism.- Each initiator shall be tested for excessive residual magnetism. Failure of any initiator to comply with the requirements of 3.3 shall be cause for rejection of the initiator.

4.2.3.3 Functional tests.- The following tests shall be performed by the Government.

4.2.3.3.1 Ballistics.- Thirty units shall be tested to determine compliance with the requirements of 3.4.1, 3.4.2, 3.4.3 and 3.4.4. The units shall be temperature conditioned as provided in Table III and 4.3.3.1.1. Failure of any unit to comply with the requirements of 3.4.1, 3.4.3 and 3.4.4 shall be cause for rejection of the lot.

TABLE III

Temperature	Sample Size
-65° to -70°F	10
65° to 75°F	10
160° to 165°F	10

4.2.3.3.2 Carbon monoxide gas leakage.- Fifteen units shall be tested to determine compliance with 3.4.2. The units shall be temperature conditioned as provided in Table III and 4.3.3.1.1. Failure of any unit to comply with the requirement of 3.4.2 shall be cause for rejection of the lot.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10522664.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench, set at the torque requirement specified in 3.2.3 shall be used in conjunction with a fixture to determine the adequacy of the joints. The body shall be secured in an air vise and the torque shall be applied to the head and the cap.

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4.3.2 Residual magnetism.- The initiator shall be tested for excessive residual magnetism, in an area free of local magnetic effects, by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.3.3 Functional.

4.3.3.1 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample unit shall be reconditioned in accordance with 4.3.3.1.1.

4.3.3.1.1 Temperature conditioning.- A standard M80 Initiator loaded with a dummy M69 cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.3.3.2 Carbon monoxide gas leakage.- The test sample shall be screwed securely into the threaded hole of the fixture base plate. A length of flared stainless steel tubing shall be attached to the initiator outlet. A length of MS28741-4 flexible hose, closed at one end with a locked shut cap shall be connected to the stainless steel tubing. The combined volumes of the tubing and flexible hose shall simulate the functional volume specified in 3.4.2. The stainless steel tubing length shall be such that the connection between the tubing and the flexible hose will be isolated from the bell jar interior. One connection only shall be permitted in the interior of the bell jar. An M3 Initiator shall be connected with an M80 Initiator simulated system hose length to the inlet port. The bell jar shall be positioned and sealed to the baseplate. A protective wire mesh cover shall be placed over the bell jar. The total set-up time shall not exceed 10 minutes, maximum before firing. Immediately prior to firing, the three way pet cock shall be aligned to assure a direct flow into the rubber bulb. After firing, the bulb shall be compressed and released several times to assure positive circulation of all gases within the bell jar. At ten minutes after each firing, the pet-cock shall be re-oriented to provide alignment with the M5A Dosimeter for carbon monoxide measurement. This procedure shall be repeated at 60 minutes after each firing. A gas sample shall be withdrawn for chromatograph analysis from the first, third and fifth unit at each temperature at both 10 and 60 minutes after firing. In the event that the carbon monoxide leakage of the second or fourth unit exceeds the maximum scale of the M5A Dosimeter, a gas sample shall be withdrawn at 60 minutes after firing for chromatograph analysis.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8797660 and C8797661.

5.2 Data card.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part the occurrence of which is other than a design function.

6.2 "Ignition delay" is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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